1.

$$\int_1^2 e^x \sin(4x)\,dx \quad , h=0.1$$

X_0	1	-2.057
X_1	1.1	-2.859
X_2	1.2	-3.3674
X_3	1.3	-3.2417
X_4	1.4	-2.5600
X_5	1.5	-1.2523
X_6	1.6	-0.5773
X_7	1.7	2.7048
X_8	1.8	4.8014
X_9	1.9	6.4714
X_{10}	2.0	7.3104

(a) Trapezoidal

$$\int_a^b f(x)dx = \frac{h}{2}\left[f(a) + 2\sum_{i=1}^{2n-1} f(x_i) + f(b)\right] = 0.39615 \ \#$$

(b) Simpson's.

$$\int_a^b f(x)dx = \frac{h}{3}\left(f(x_0) + 2\sum_{i=2}^{n} f(x_{2i-2}) + 4\sum_{i=1}^{n} f(x_{2i-1}) + f(x_{2n})\right)$$

$$= 0.385664 \ \#$$

(c) midpoint. $\int_a^b f(x)dx = 2h\left[f(x_1) + f(x_3) + \dots f(x_{2n-1})\right] = 0.36470 \ \#$

2. $\int_1^{1.5} x^2 \ln x\, dx$

$a=1 \quad b=1.5$

$n=3 \Rightarrow \left| \begin{array}{l} \eta_1 = \sqrt{\frac{3}{5}} \\ \eta_2 = 0 \\ \eta_3 = -\sqrt{\frac{3}{5}} \end{array} \right.$ $\quad c_1 = \frac{5}{9} = c_3$, $c_2 = \frac{8}{9}$.

$x = \frac{b-a}{2}\eta + \frac{a+b}{2}$

$= \frac{1}{4}\eta + \frac{5}{4}$

$dx = \frac{1}{4}d\eta$

$\Rightarrow X_1 = 1.0564$

$X_2 = \frac{5}{4}$

$X_3 = 1.4436$

$\Rightarrow \frac{1.5-1}{2}\sum_{i=1}^{3} c_i f(x_i) = 0.19226 \ \#$

$n=4 \Rightarrow x_1 = 1.0347$, $x_2 = 1.1650$, $x_3 = 1.3350$, $x_4 = 1.4653$

$C_1 = \frac{18-\sqrt{30}}{36} = C_4$

$C_2 = \frac{18+\sqrt{30}}{36} = C_3$

$$\Rightarrow \frac{1}{4}\sum_{i=1}^{4} c_i f(x_i) = 0.19226 \ \#$$

3. by 1、2 的重積分.

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Problem 3 Results:
a. Simpson's rule (n=4, m=4): 0.51198754
b. Gaussian Quadrature (n=3, m=3): 0.51186554
c. Exact value: 0.51184464
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4. (a) Calculate by python

(b) $\int_1^{\infty} x^{-4} \sin dx$ improper integral Transform $t = x^{-1}$

$dx = \frac{-1}{t^2} dt$

$$\int_1^0 t^4 \sin\frac{1}{t}\left(-\frac{1}{t^2}dt\right) = \int_0^1 t^2 \sin\left(\frac{1}{t}\right) dt$$

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Problem 4 Results:
a. ∫₀¹ x^(-1/4) sin(x) dx using composite Simpson's rule (n=4): 0.52609449
b. ∫₁^∞ x^(-4) sin(x) dx using transformed composite Simpson's rule (n=4): 0.27448161
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